

Chapter → 12

Packages

Interpreter vs Compiler

Interpreter translates one statement at a time into machine code.

Compiler scans the entire program and translates whole of it into machine code.

Interpreter

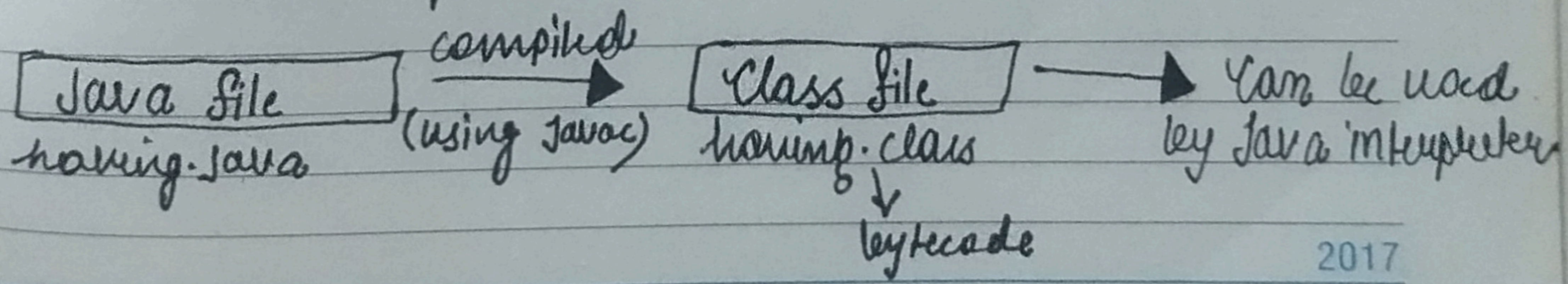
vs

Compiler

- | | |
|------------------------------------|---|
| ★ One Statement at a time. | ★ Entire Program at a time |
| ★ Interpreter is needed everytime. | ★ Once compiled, it is not needed. |
| ★ Partial execution if error | ★ No execution, if error occurs. |
| ★ Easy for programmers | ★ Usually not as easy as interpreted one. |

Is Java compiled or interpreter?

Java is a hybrid language → both compiled as well as interpreted. SUN 16



- A JVM can be used to interpret this bytecode.
- This bytecode can be taken to any platform (win/mac/linux) for execution.
- Hence Java is platform independent (write once run everywhere).

12 Executing a Java Program

1 `Javac Harry.java` → Compiled
 2 `Javac Harry.class` → Interpreted

3 So far the execution of our program was being managed by intelliJ idea.

4 we can download a source code editor like VS code to compile & execute our Java program.

5

6

7

Packages in Java

A Package is used to group related classes. Packages help in avoiding name conflicts.

There are two types of Packages:

- * Built In Packages → Java API
- * User defined Packages → Custom Packages

Creating a package

`javac Hawvy.java` → Creates Hawvy.class

`javac -d . Hawvy.java` → Creates a package folder
↳ we can keep adding classes to a package like this.

We can also create inner packages by adding "Package.inner" as package name.
↓
Folder Subfolder

These packages once created can be used by other classes.

Access Modifiers in Java

- ¹⁰ Access modifiers determine whether other classes can be use a particular field or invoke a particular method
- ¹¹ can be public, private, protected or default (no modifier)

12

Modifiers	class	Package	Subclass	module
¹ public	Y	Y	Y	Y
protected	Y	Y	Y	N
² Default	Y	Y	N	N
private	Y	N	N	N

3

4

5

6

7